

# Enhanced Correlation Coefficient as a Refinement of Image Registration

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**Abstract**—A study of the effectiveness of Enhanced Correlation Coefficient (ECC) on the performance of feature-based image registration approaches is carried out. This investigation determines if ECC improves image registration performance on datasets which test on invariance to scale, rotation and viewpoint change. Five state-of-the-arts methods are considered, namely KAZE, Binary Robust Invariant Scalable Keypoints (BRISK), Oriented FAST and Rotated Brief (ORB), Speeded-Up Robust Features (SURF), and Scale-Invariant Feature Transform (SIFT). Root-mean-squared error of control points is used to evaluate the image registration performance on datasets taken from the Oxford Robotics Database. A global ranking factor is used to rank each method within a dataset. The efficiency of each method is recorded as a guide for selecting a method for a specific application. Results indicate that ECC improves image registration performance in most cases with a small time addition.

## I. INTRODUCTION

Image registration is the alignment of a target image to the coordinate system of a reference image. Its major steps are feature detection, feature description, feature matching, Random Sample Consensus (RANSAC) [1], and spatial transformation of the target image.

There are two types of registration: feature-based and area-based [2]. Feature-based image registration involves detection and description of interest points within the reference image and target image. It is suitable for cases where there is minimal overlap between the two images. However, it tends to fail when there is repeated patterns within the images.

Area-based image registration aligns two images based on a similarity measure using pixel intensity values. It aligns images to a high degree of accuracy because it uses information from the whole image [3]. It requires a good initialization to avoid from being trapped in local extrema.

Image registration has wide applications including in remote sensing, the medical field, 3D reconstruction, and motion estimation of autonomous vehicles equipped with cameras. This paper focusses on its applications in outdoor images which simulate challenges faced by autonomous vehicles.

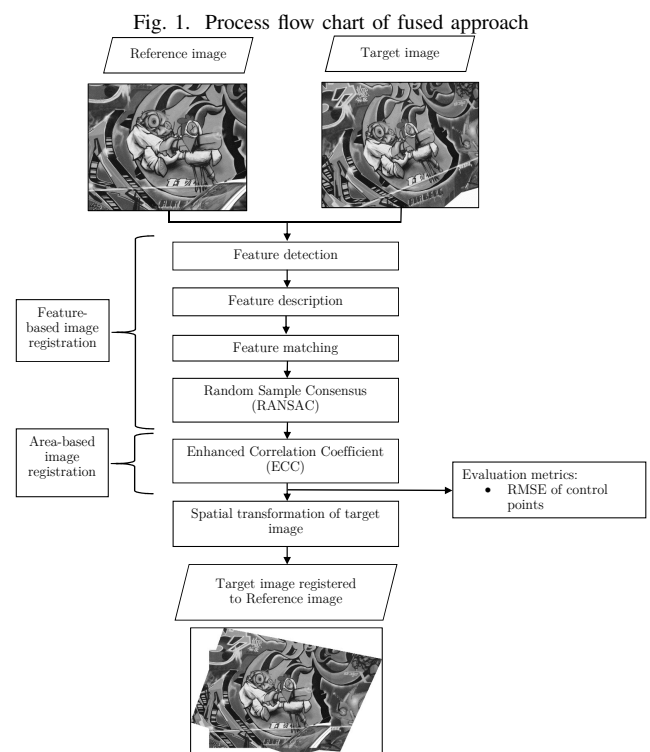
Szeliski suggested that a more accurate transformation matrix can be estimated using area-based alignment after initial alignment by a feature-based approach. The dense nature of area-based approaches allows it to handle parallax well [4]. Evangelidis's Image Alignment Toolbox (IAT) [5]

implemented this approach by using Speeded-Up Robust Features (SURF) [6] as the feature-based method and ECC [7] as the area-based approach with favourable results for mosaic creation.

The objective of this paper is to investigate the effects of ECC as a refinement for image registration. It uses five state-of-the-art feature-based methods, namely KAZE [8], Binary Robust Invariant Scalable Keypoints (BRISK) [9], Oriented FAST and Rotated Brief (ORB) [10], Speeded-Up Robust Features (SURF), and Scale-Invariant Feature Transform (SIFT) [11].

## II. METHODOLOGY

The following sections describe the image registration process and ECC's role. In the feature detection and description section, each type of feature that is being considered is examined. Similarly, the formulation of the feature matching, RANSAC, ECC, and spatial transformation stage are explained in detail. Fig. 1 illustrates the whole image registration process.



### A. Feature detection and description

1) *KAZE*: This feature respects natural boundaries in images, unlike previous methods which apply Gaussian filtering uniformly on the whole image. It does this by utilizing a nonlinear scale space through Additive Operator Splitting and variable conductance diffusion. Hence, its filter is locally adaptive which results in higher localization and distinctiveness in features. It has comparable complexity with SIFT and higher complexity than SURF.

2) *BRISK*: BRISK uses intra-octaves in addition to octaves to speed up interest point detection. It fits a quadratic function to extrapolate an interest point's location and scale.

Its descriptor is binary. It has a hand-crafted sampling pattern with varying Gaussian filtering. It has two types of sampling pairs, namely short and long pairs. Short pairs are used for intensity comparisons to represent the descriptor. Long pairs are used to compute the orientation. The local gradient,  $g(p_i, p_j)$ , between sampling pairs,  $(p_i, p_j)$ , is defined as:

$$g(p_i, p_j) = (p_j - p_i) \cdot \frac{I(p_j, \sigma_j) - I(p_i, \sigma_i)}{\|p_j - p_i\|^2} \quad (1)$$

where  $I(p_i, \sigma_i)$  is the pixel intensity of the sampling point smoothed by the corresponding Gaussian filtering level. All the local gradients are added up. The orientation  $\theta = \tan^{-1}(g_y/g_x)$  where  $g_x$  and  $g_y$  are the  $x$  and  $y$  components of the gradient respectively.

3) *ORB*: Oriented FAST and Rotated Brief (ORB) uses FAST[12] with a circular radius of 9 pixels to detect features at each level of a scale pyramid. The Harris corner measure is used to rank a target number of keypoints. The orientation of each keypoint is assigned using the intensity centroid method which uses a vector between a corner and its center to represent its orientation.

ORB is similar to the Binary Robust Independent Elementary Features (BRIEF) descriptor [13] in that it does not have an elaborate sampling pattern. It learns its sampling pattern by maximising uncorrelation and variance among sampling pairs. It takes a training set of 300,000 keypoints from the PASCAL 2006 dataset [14]. Its algorithm first runs each sampling pair out of a possible 205,000 against all patches. Then, a list of pairs ordered by distance from mean of 0.5 is stored in a vector,  $T$ . It performs its search by moving the first pair from  $T$  into a result vector,  $R$ . After that, it moves the next pair from  $T$  to  $R$ . If the pair's correlation with all pairs in  $R$  is above a threshold,  $t_c$ , it is discarded. Otherwise, the pair is kept. This step is repeated until 256 pairs are reached in  $R$ . If the number of pairs is less than 256, the correlation threshold is raised and the search is redone.

4) *SURF*: SURF approximates the Laplacian of Gaussian (LoG) in the scale pyramid with a box filter. It detects blobs using the Determinant of Hessian (DoH). For orientation assignment, it computes Gaussian-weighted wavelet responses in a  $6s \times 6s$  neighbourhood with integral images, where  $s$  is a predefined number of pixels. The wavelet response is plotted in a graph where the largest sum of responses in a sliding  $60^\circ$  window is defined as the orientation.

It describes interest points by taking a  $20s \times 20s$  neighbourhood and dividing it into  $4 \times 4$  subregions. A four-element vector,  $v$ , is computed in each subregion as  $v = (\sum d_x, \sum d_y, \sum |d_x|, \sum |d_y|)$  where  $d_x$  and  $d_y$  are pixel intensity derivatives in the  $x$  and  $y$  axis respectively.

5) *SIFT*: The Scale-Invariant Feature Transform (SIFT) detector uses Difference of Gaussians (DoG) as an approximation of Laplacian of Gaussians (LoG) blob detector in a Gaussian pyramid. Then, scale-space extrema are detected in these DoGs. A contrast threshold is applied on extrema locations refined by Taylor's series expansion. In addition, an edge threshold is applied on a ratio of eigenvalues obtained from a Hessian matrix. For orientation assignment, a neighbourhood around the keypoint is taken based on the scale. Then, the region's gradient magnitude and direction is computed. The top 20% of peaks in a 36-bin histogram covering 360 degrees are considered for orientation calculation.

SIFT features are described by taking a  $16 \times 16$  neighbourhood around the keypoint. This neighbourhood is divided into 16 sub-blocks of  $4 \times 4$  size. An 8 bin orientation histogram is created for each sub-block.

### B. Feature matching

The distance between feature descriptors is computed through exhaustive search of their sum-of-squared differences (SSD). A threshold is applied on the distance ratio of the features' SSDs to avoid ambiguous matches which are too close to each other. A matching threshold is applied on the matches to remove false positives.

For binary descriptors like BRISK and ORB, the distance between them is defined as the total number of different bits. This is done by performing an *XOR* operation on both descriptors and summing up the result.

### C. RANSAC

Feature matches are used by RANSAC to produce inliers and the initial transformation matrix. If the fraction of inliers to total matches exceeds a threshold, the transformation matrix is estimated using the inliers and this process terminates.

### D. ECC

ECC is a modified version of correlation coefficient which has the added benefit of invariance to photometric distortions. Its performance criterion is:

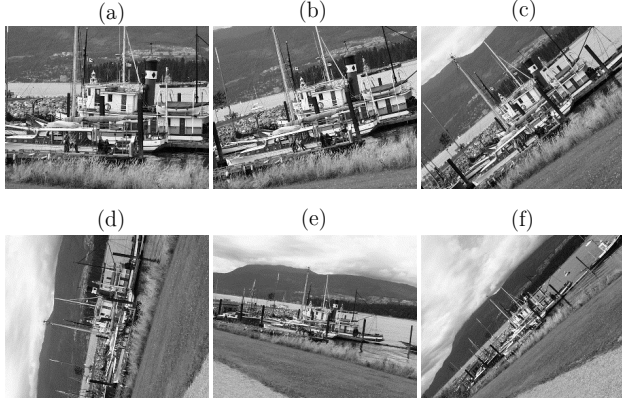
$$E_{ECC(p)} = \left\| \frac{\bar{i}_r}{\|\bar{i}_r\|} - \frac{\bar{i}_w(p)}{\|\bar{i}_w(p)\|} \right\|^2 \quad (2)$$

where  $p$  is the transformation parameters,  $\bar{i}_r$  is the zero-mean of the reference image vector, and  $\bar{i}_w$  is the zero-mean of the warped image vector.

Eq.2 indicates that ECC is invariant to bias and gain changes which translates to brightness and contrast invariance.

A forward additive and an inverse compositional approach are developed to maximize ECC's performance criterion. These iterative schemes approximate the nonlinear objective function to produce a closed-form solution per iteration.

Fig. 2. Images from the Boat dataset. (a) is the reference image. (b) - (f) are target images that are to be aligned to the reference image.



### E. Spatial transformation of target image

This process involves warping the coordinate system of the target image. Homography is employed as the transformation model. The target image coordinates are converted to homogeneous coordinates:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (3)$$

where  $x, y$  are the initial coordinates,  $u, v$  are the final coordinates, and  $a - h$  are homography model parameters which describe translation, scaling, and rotation.

In conclusion, a target and a reference image are taken as input into the fused approach. Firstly, features are detected and described in both images. Then, feature descriptors are matched. Feature matches are used as input to RANSAC which produces the inliers and initial transformation matrix. The initial transformation matrix is inputted to ECC for refinement. Finally, the resulting transformation matrix is used to warp the target image, registering it to the reference image.

## III. EXPERIMENTS

The setup of the experiments in terms of the dataset, evaluation metrics, and implementation details is discussed. After that, the experimental results are analyzed in both the Boat and Graffiti datasets.

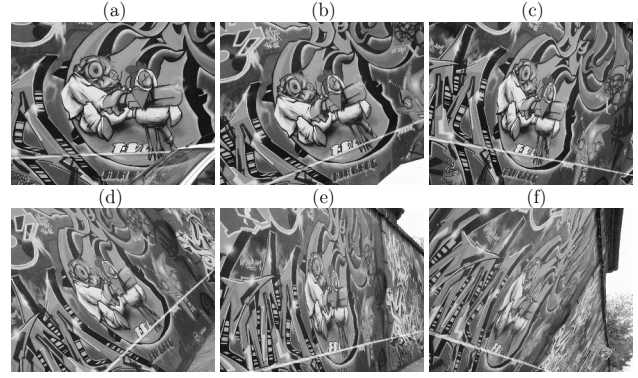
### A. Setup

1) *Dataset*: The Oxford Robotics Database[15] is used. This database is the standard database used for evaluating image registration performance. The Boat and Graffiti datasets are used for experiments.

Fig.2 shows the Boat dataset. It consists of images of boats which are rotated and scaled to varying degrees. The Graffiti dataset consists of images of graffiti from various viewpoints as shown in Fig.3.

In both datasets, the first image is considered the reference image. The remaining five images are treated as target images to be registered to the reference image.

Fig. 3. Images from the Graffiti dataset. (a) is the reference image. (b) - (f) are target images that are to be aligned to the reference image.



2) *Metrics*: The root-mean-squared error of control points [16], [17],  $rmse_{cp}$ , is used to evaluate the image registration performance between two images. It is defined as:

$$rmse_{cp} = \sqrt{\frac{\sum_{i=1}^n e_i^2}{n}} \quad (4)$$

where  $n$  is the number of corresponding control points and  $e$  is the Euclidean distance error of matching points. To avoid ambiguity in results, a global metric, known as the ranking factor,  $R$ , is used to represent the performance of each feature in each experiment. It is defined as  $R_{Boat/Graffiti} = (10 - rmse_{cp_{avg}}) \times n_{rmse_{cp}}$  where  $R_{Boat}$  and  $R_{Graffiti}$  are the ranking factors for the Boat and Graffiti dataset respectively. 10 is empirically defined as the largest accepted rmse tolerance.  $rmse_{cp_{avg}}$  represents the average  $rmse_{cp}$  across all target images within each dataset.  $n_{rmse_{cp}}$  is the number of target images that are aligned within the accepted rmse tolerance of 10 for each dataset.

The time taken for the feature-based methods and ECC is recorded to compare the efficiency of various combinations of methods.  $t_{fb}$  is the time taken to register the images using the corresponding feature-based method.  $t_{ab}$  is the time taken to refine the registration using an area-based method. In this case, ECC is used as the area-based method.  $t_T$  is the total time taken for the fused approach which is the sum of  $t_{fb}$  and  $t_{ab}$ .

3) *Implementation*: The fused approach is based on a demonstration in IAT. OpenCV's implementations of features, RANSAC and ECC are used to produce the matching feature points, inliers and transformation matrices. Default parameter values are used for these implementations.  $rmse_{cp}$  is implemented in MATLAB to evaluate the image registration performance.

### B. Results

1) *Boat dataset*: Table I indicates that KAZE performs better after ECC ( $R_{Boat} = 46.6140$ ) than before ECC ( $R_{Boat} = 43.4993$ ). This largely due to ECC's significant effect on the alignment of img6 with an  $rmse_{cp}$  improvement of 3.3599 pixels. This shows that ECC helps to compensate for KAZE's drawbacks in handling large scale and rotation change.

In Table II, BRISK's performance before ( $R_{Boat} = 38.8407$ ) and after ECC ( $R_{Boat} = 38.7841$ ) is almost the

TABLE I

PERFORMANCE OF KAZE ON THE BOAT DATASET.  $kaze_{before}$  AND  $kaze_{after}$  MEAN KAZE BEFORE AND AFTER ECC RESPECTIVELY. THE MEANING IS THE SAME IN OTHER TABLES CORRESPONDING TO THE FEATURE-BASED METHOD.

| Target image    | $kaze_{before}$ |          | $kaze_{after}$ |          |         |
|-----------------|-----------------|----------|----------------|----------|---------|
|                 | $rmse_{cp}$     | $t_{fb}$ | $rmse_{cp}$    | $t_{ab}$ | $t_T$   |
| img2            | 0.1105          | 50.6771  | 0.1819         | 0.7513   | 51.4284 |
| img3            | 0.1445          | 45.2231  | 0.0948         | 0.7541   | 45.9772 |
| img4            | 0.4370          | 40.4955  | 0.5323         | 1.2563   | 41.7518 |
| img5            | 0.2712          | 39.9109  | 0.3993         | 1.2437   | 41.1546 |
| img6            | 5.5375          | 40.1459  | 2.1776         | 2.6767   | 42.8226 |
| Average         | 1.3001          | 43.2905  | 0.6772         | 1.3364   | 44.6269 |
| $n_{rmse_{cp}}$ | 5               | 5        | 5              | 5        | 5       |
| $R_{Boat}$      | 43.4993         |          | 46.6140        |          |         |

TABLE II

PERFORMANCE OF BRISK ON THE BOAT DATASET. THE SYMBOL ‘-’ MEANS THAT THE METHOD FAILED FOR THE CORRESPONDING TARGET IMAGE. THE MEANING OF THIS SYMBOL IN OTHER TABLES IS THE SAME.

| Target image    | $brisk_{before}$ |          | $brisk_{after}$ |          |         |
|-----------------|------------------|----------|-----------------|----------|---------|
|                 | $rmse_{cp}$      | $t_{fb}$ | $rmse_{cp}$     | $t_{ab}$ | $t_T$   |
| img2            | 0.1122           | 27.3375  | 0.1819          | 0.7427   | 28.0802 |
| img3            | 0.2186           | 23.6714  | 0.0964          | 0.7558   | 24.4272 |
| img4            | 0.4200           | 18.5942  | 0.5345          | 1.6391   | 20.2333 |
| img5            | 0.4084           | 16.5670  | 0.4030          | 1.4101   | 17.9771 |
| img6            | -                | -        | -               | -        | -       |
| Average         | 0.2898           | 21.5425  | 0.3040          | 1.1369   | 22.6794 |
| $n_{rmse_{cp}}$ | 4                | 4        | 4               | 4        | 4       |
| $R_{Boat}$      | 38.8407          |          | 38.7841         |          |         |

same. This is because BRISK already has implementations to handle large scale and rotation changes on its own. Its use of long sampling pairs handles rotation. Besides that, its use of intra-octaves to extrapolate the scale of interest points helps it to handle magnification.

Table III indicates that ORB performs better after ECC ( $R_{Boat} = 29.3160$ ) than before it ( $R_{Boat} = 27.5424$ ). This is due to  $rmse_{cp}$  improvements in img2, img3, and img5. ORB’s lower performance compared to other features indicates low robustness although it offers real-time efficiency. As a result, it fails for two images (img4 and img6). This means that there is no result after ECC for these images because ECC depends on a good initialization.

Table IV shows that SURF performs better after ECC ( $R_{Boat} = 38.7890$ ) than before ECC ( $R_{Boat} = 38.6296$ ). It fails for img6 because its box-filtered implementation is not able to handle large scale and rotation change. For img4 and img5, the  $rmse_{cp}$  increased after ECC. This shows that ECC has reached maximum convergence because SURF has already handled rotation and scale well with its wavelet response and approximated scale pyramid

TABLE III

PERFORMANCE OF ORB ON THE BOAT DATASET

| Target image    | $orb_{before}$ |          | $orb_{after}$ |          |        |
|-----------------|----------------|----------|---------------|----------|--------|
|                 | $rmse_{cp}$    | $t_{fb}$ | $rmse_{cp}$   | $t_{ab}$ | $t_T$  |
| img2            | 0.5580         | 2.1132   | 0.1820        | 0.9781   | 3.0913 |
| img3            | 0.3447         | 1.8236   | 0.0948        | 0.9555   | 2.7792 |
| img4            | -              | -        | -             | -        | -      |
| img5            | 1.5548         | 1.8731   | 0.4072        | 1.3632   | 3.2362 |
| img6            | -              | -        | -             | -        | -      |
| Average         | 0.8192         | 1.9366   | 0.2280        | 1.0989   | 3.0356 |
| $n_{rmse_{cp}}$ | 3              | 3        | 3             | 3        | 3      |
| $R_{Boat}$      | 27.5424        |          | 29.3160       |          |        |

TABLE IV

PERFORMANCE OF SURF ON THE BOAT DATASET

| Target image    | $surf_{before}$ |          | $surf_{after}$ |          |         |
|-----------------|-----------------|----------|----------------|----------|---------|
|                 | $rmse_{cp}$     | $t_{fb}$ | $rmse_{cp}$    | $t_{ab}$ | $t_T$   |
| img2            | 0.2992          | 15.3138  | 0.1821         | 0.9711   | 16.2849 |
| img3            | 0.3939          | 12.3229  | 0.0950         | 0.9648   | 13.2877 |
| img4            | 0.3956          | 11.6725  | 0.5353         | 1.2040   | 12.8765 |
| img5            | 0.2817          | 12.9344  | 0.3986         | 0.9951   | 13.9295 |
| img6            | -               | -        | -              | -        | -       |
| Average         | 0.3426          | 13.0609  | 0.3028         | 1.0337   | 14.0946 |
| $n_{rmse_{cp}}$ | 4               | 4        | 4              | 4        | 4       |
| $R_{Boat}$      | 38.6296         |          | 38.7890        |          |         |

TABLE V

PERFORMANCE OF SIFT ON THE BOAT DATASET

| Target image    | $sift_{before}$ |          | $sift_{after}$ |          |         |
|-----------------|-----------------|----------|----------------|----------|---------|
|                 | $rmse_{cp}$     | $t_{fb}$ | $rmse_{cp}$    | $t_{ab}$ | $t_T$   |
| img2            | 0.3084          | 28.3282  | 0.1819         | 1.0069   | 29.3351 |
| img3            | 0.2550          | 24.3930  | 0.0957         | 1.0954   | 25.4884 |
| img4            | 0.4939          | 22.0328  | 0.5343         | 1.2196   | 23.2524 |
| img5            | 0.3648          | 20.6773  | 0.4000         | 1.1201   | 21.7974 |
| img6            | 2.0361          | 20.8208  | 2.1792         | 1.3586   | 22.1794 |
| Average         | 0.6916          | 23.2504  | 0.6782         | 1.1601   | 24.4105 |
| $n_{rmse_{cp}}$ | 5               | 5        | 5              | 5        | 5       |
| $R_{Boat}$      | 46.5419         |          | 46.6089        |          |         |

approach respectively. Overall, the  $rmse_{cp}$  improved by 0.1593 pixel across all target images.

According to Table V, SIFT has a higher  $R_{Boat}$  after ECC ( $R_{Boat} = 46.6089$ ) than before ECC ( $R_{Boat} = 46.5419$ ). Its DoG approach in the scale pyramid and its histogram approach to orientation performs well in this dataset. The  $rmse_{cp}$  decreased after ECC for img4, img5, and img6 because ECC has maximized its convergence. However, the  $rmse_{cp}$  across all target images increases by 0.0670 pixel.

Table VI indicates that all feature-based methods except BRISK are enhanced by ECC with a small time addition. ORB’s performance is increased with a time difference of 1.099 seconds. Similarly, SURF performs better with an addition of 1.0377 seconds. BRISK’s  $R_{Boat}$  decrease of 0.056 is still lower than the overall minimum ranking factor increase of 0.0670 in SIFT which has a time addition of 1.1601 seconds. KAZE’s performance is increased to a level of accuracy equivalent to SIFT.

2) *Graffiti dataset*: KAZE performs much better after ECC ( $R_{Boat} = 28.2572$ ) than before ECC ( $R_{Boat} = 23.1482$ ) based on Table VII. ECC’s positive effect increases exponentially as the viewpoint change increases. This shows that ECC is suitable for refining registrations

TABLE VI

$R_{Boat}$  AND AVERAGE TIME TAKEN FOR EACH METHOD IN THE BOAT DATASET

| Method           | $R_{Boat}$ | Average time taken |
|------------------|------------|--------------------|
| $orb_{before}$   | 27.5424    | 1.9366             |
| $orb_{after}$    | 29.3160    | 3.0356             |
| $surf_{before}$  | 38.6296    | 13.0609            |
| $surf_{after}$   | 38.7890    | 14.0946            |
| $brisk_{before}$ | 38.8407    | 21.5425            |
| $brisk_{after}$  | 38.7841    | 22.6794            |
| $sift_{before}$  | 46.5419    | 23.2504            |
| $sift_{after}$   | 46.6089    | 24.4105            |
| $kaze_{before}$  | 43.4993    | 43.2905            |
| $kaze_{after}$   | 46.6140    | 44.6269            |

TABLE VII  
PERFORMANCE OF KAZE ON THE GRAFFITI DATASET

| Target image                | <i>kaze<sub>before</sub></i> |                       | <i>kaze<sub>after</sub></i> |                       |                      |
|-----------------------------|------------------------------|-----------------------|-----------------------------|-----------------------|----------------------|
|                             | <i>rmse<sub>cp</sub></i>     | <i>t<sub>fb</sub></i> | <i>rmse<sub>cp</sub></i>    | <i>t<sub>ab</sub></i> | <i>t<sub>T</sub></i> |
| img2                        | 0.4603                       | 34.3162               | 0.5799                      | 1.1414                | 35.4576              |
| img3                        | 1.8182                       | 33.9647               | 0.6141                      | 1.2493                | 35.2140              |
| img4                        | 4.5732                       | 32.7232               | 0.5487                      | 3.2912                | 36.0144              |
| img5                        | -                            | -                     | -                           | -                     | -                    |
| img6                        | -                            | -                     | -                           | -                     | -                    |
| Average                     | 2.2839                       | 33.6680               | 0.5809                      | 1.8939                | 35.5620              |
| <i>nrmse<sub>cp</sub></i>   | 3                            | 3                     | 3                           | 3                     | 3                    |
| <i>R<sub>Graffiti</sub></i> | 23.1483                      |                       | <b>28.2572</b>              |                       |                      |

TABLE VIII  
PERFORMANCE OF BRISK ON THE GRAFFITI DATASET

| Target image                | <i>brisk<sub>before</sub></i> |                       | <i>brisk<sub>after</sub></i> |                       |                      |
|-----------------------------|-------------------------------|-----------------------|------------------------------|-----------------------|----------------------|
|                             | <i>rmse<sub>cp</sub></i>      | <i>t<sub>fb</sub></i> | <i>rmse<sub>cp</sub></i>     | <i>t<sub>ab</sub></i> | <i>t<sub>T</sub></i> |
| img2                        | 0.4318                        | 10.7755               | 0.5807                       | 1.0085                | 11.7840              |
| img3                        | 1.5278                        | 10.8338               | 0.5717                       | 1.1956                | 12.0294              |
| img4                        | -                             | -                     | -                            | -                     | -                    |
| img5                        | -                             | -                     | -                            | -                     | -                    |
| img6                        | -                             | -                     | -                            | -                     | -                    |
| Average                     | 0.9798                        | 10.8047               | 0.5762                       | 1.1020                | 11.9067              |
| <i>nrmse<sub>cp</sub></i>   | 2                             | 2                     | 2                            | 2                     | 2                    |
| <i>R<sub>Graffiti</sub></i> | 18.0404                       |                       | <b>18.8476</b>               |                       |                      |

when target images have large viewpoint change.

Table VIII indicates that BRISK has a higher  $R_{Graffiti}$  after ECC ( $R_{Graffiti} = 18.8476$ ) than before ECC ( $R_{Graffiti} = 18.0404$ ). BRISK's performance for img2 decreases after ECC because BRISK already handles small viewpoint change well. It shows ECC's tendency to converge to an  $rmse_{cp}$  value of 0.5 pixel. The  $rmse_{cp}$  improvement for img3 after ECC enforces ECC's enhancement pattern for images with large viewpoint change. BRISK's failure to register img4, img5, and img6 indicates its low tolerance to viewpoint change because it has little implementation to handle this type of distortion.

ORB's performance improves after ECC ( $R_{Graffiti} = 18.7933$ ) compared to before ECC ( $R_{Graffiti} = 11.5779$ ) according to Table IX. ECC's enhancement is more pronounced as the viewpoint change increases. For img3, the  $rmse_{cp}$  improvement is 6.249 pixels. ORB's failure for img4, img5, img6 indicates that its low tolerance to large viewpoint change.

Table X shows that SURF's  $R_{Graffiti}$  after ECC ( $R_{Graffiti} = 18.8361$ ) is higher than before ECC ( $R_{Graffiti} = 12.9189$ ). It improves the registration of both img2 and img3 to subpixel accuracy (Average  $rmse_{cp} = 0.5819$  pixel). SURF's failure for img4, img5, and img6 shows that it has no specific implementation to handle large

TABLE IX  
PERFORMANCE OF ORB ON THE GRAFFITI DATASET

| Target image                | <i>orb<sub>before</sub></i> |                       | <i>orb<sub>after</sub></i> |                       |                      |
|-----------------------------|-----------------------------|-----------------------|----------------------------|-----------------------|----------------------|
|                             | <i>rmse<sub>cp</sub></i>    | <i>t<sub>fb</sub></i> | <i>rmse<sub>cp</sub></i>   | <i>t<sub>ab</sub></i> | <i>t<sub>T</sub></i> |
| img2                        | 1.5473                      | 1.7760                | 0.5809                     | 1.3792                | 3.1552               |
| img3                        | 6.8748                      | 1.8715                | 0.6258                     | 1.7357                | 3.6072               |
| img4                        | -                           | -                     | -                          | -                     | -                    |
| img5                        | -                           | -                     | -                          | -                     | -                    |
| img6                        | -                           | -                     | -                          | -                     | -                    |
| Average                     | 4.2111                      | 1.8237                | 0.6033                     | 1.5574                | 3.3812               |
| <i>nrmse<sub>cp</sub></i>   | 2                           | 2                     | 2                          | 2                     | 2                    |
| <i>R<sub>Graffiti</sub></i> | 11.5779                     |                       | <b>18.7933</b>             |                       |                      |

TABLE X  
PERFORMANCE OF SURF ON THE GRAFFITI DATASET

| Target image                | <i>surf<sub>before</sub></i> |                       | <i>surf<sub>after</sub></i> |                       |                      |
|-----------------------------|------------------------------|-----------------------|-----------------------------|-----------------------|----------------------|
|                             | <i>rmse<sub>cp</sub></i>     | <i>t<sub>fb</sub></i> | <i>rmse<sub>cp</sub></i>    | <i>t<sub>ab</sub></i> | <i>t<sub>T</sub></i> |
| img2                        | 1.6350                       | 10.0103               | 0.5866                      | 1.0502                | 11.0605              |
| img3                        | 5.4461                       | 10.6119               | 0.5772                      | 3.4597                | 14.0716              |
| img4                        | -                            | -                     | -                           | -                     | -                    |
| img5                        | -                            | -                     | -                           | -                     | -                    |
| img6                        | -                            | -                     | -                           | -                     | -                    |
| Average                     | 3.5406                       | 10.3111               | 0.5819                      | 2.2550                | 12.5661              |
| <i>nrmse<sub>cp</sub></i>   | 2                            | 2                     | 2                           | 2                     | 2                    |
| <i>R<sub>Graffiti</sub></i> | 12.9189                      |                       | <b>18.8361</b>              |                       |                      |

TABLE XI  
PERFORMANCE OF SIFT ON THE GRAFFITI DATASET

| Target image                | <i>sift<sub>before</sub></i> |                       | <i>sift<sub>after</sub></i> |                       |                      |
|-----------------------------|------------------------------|-----------------------|-----------------------------|-----------------------|----------------------|
|                             | <i>rmse<sub>cp</sub></i>     | <i>t<sub>fb</sub></i> | <i>rmse<sub>cp</sub></i>    | <i>t<sub>ab</sub></i> | <i>t<sub>T</sub></i> |
| img2                        | 0.4411                       | 10.3365               | 0.5787                      | 0.8499                | 11.1864              |
| img3                        | 2.2922                       | 10.8907               | 0.6210                      | 1.5444                | 12.4351              |
| img4                        | -                            | -                     | -                           | -                     | -                    |
| img5                        | -                            | -                     | -                           | -                     | -                    |
| img6                        | -                            | -                     | -                           | -                     | -                    |
| Average                     | 1.3666                       | 10.6136               | 0.5998                      | 1.1971                | 11.8107              |
| <i>nrmse<sub>cp</sub></i>   | 2                            | 2                     | 2                           | 2                     | 2                    |
| <i>R<sub>Graffiti</sub></i> | 17.2668                      |                       | <b>18.8004</b>              |                       |                      |

viewpoint distortion.

Based on Table XI, SIFT performs better after ECC ( $R_{Graffiti} = 18.8004$ ) than before ECC ( $R_{Graffiti} = 17.2668$ ). This is largely due to the reduction in  $rmse_{cp}$  of 1.6712 pixels for img3. SIFT fails for img4, img5, and img6. This indicates a gap in its performance although it has high accuracy for changes in scale and rotation, and small changes in viewpoint.

The performance of some feature-based methods in the Graffiti dataset are elevated to the next level with little trade-off in efficiency according to Table XII. ORB's  $R_{Graffiti}$  increases by 7.2154 with a time addition of 1.5575 seconds. ECC's compensates for SURF's poor performance by increasing its  $R_{Graffiti}$  by 5.9172 and adding only 2.255 seconds. SIFT's and BRISK's performances are increased with additions of 1.1973 seconds and 1.102 seconds respectively. KAZE's  $R_{Graffiti}$  is increased by 5.1089 with an addition of 1.894 seconds.

Experimental results indicate that ECC enhances image registration performance consistently on most feature-based methods in the Boat dataset. In the Graffiti dataset, ECC has a large impact on the ranking factors of certain feature-based methods. In addition, this enhancement comes with only a minimal addition in operation time in the range

TABLE XII  
 $R_{Graffiti}$  AND AVERAGE TIME TAKEN FOR EACH METHOD IN THE GRAFFITI DATASET

| Method                        | $R_{Graffiti}$ | Average time taken |
|-------------------------------|----------------|--------------------|
| <i>orb<sub>before</sub></i>   | 11.5779        | 1.8237             |
| <i>orb<sub>after</sub></i>    | 18.7933        | 3.3812             |
| <i>surf<sub>before</sub></i>  | 12.9189        | 10.3111            |
| <i>sift<sub>before</sub></i>  | 17.2668        | 10.6136            |
| <i>brisk<sub>before</sub></i> | 18.0404        | 10.8047            |
| <i>sift<sub>after</sub></i>   | 18.8004        | 11.8107            |
| <i>brisk<sub>after</sub></i>  | 18.8476        | 11.9067            |
| <i>surf<sub>after</sub></i>   | 18.8361        | 12.5661            |
| <i>kaze<sub>before</sub></i>  | 23.1483        | 33.6680            |
| <i>kaze<sub>after</sub></i>   | 28.2572        | 35.5620            |

of between 0.7427 and 3.4597 seconds. The performance of each method is evaluated using a ranking factor which takes into account the  $rmse_{cp}$  of the method across all target images.

Besides mostly positive results, there are also minor decreases in  $rmse_{cp}$  scattered throughout all methods after ECC. In most of these cases, the images are already aligned to subpixel accuracy. The only exception is SIFT on img6 of the Boat dataset which may be due to complications arising from a bad initialisation. In all cases, the  $rmse_{cp}$  decrease is no larger than 0.1489 pixels. Results from the decreases in performance suggest that ECC tends to converge at an  $rmse_{cp}$  of 0.5859 pixels on average.

#### IV. CONCLUSION

The image registration performance of most feature-based methods has been improved by ECC in datasets with varying scale, rotation, and viewpoint. ORB, SURF, and KAZE have been improved by a large margin in the Graffiti dataset. In the Boat dataset, all feature-based methods are enhanced except for a slight decrease for BRISK. This increase in performance comes with only a small addition in time. The results comprising of the ranking factor and time taken serve as a guide for selection of the methods for a specific application. Potential future works include precise template matching applications in robot navigation, motion detection, and quality inspections.

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